# SUMMARY TABLE: CHEMISTRY GRADE 10

*Sub-Strand 1.2: The Atom — Teacher Reference (Pre-filled)*

| SUMMARY TABLE: CHEMISTRY GRADE 10 | |
|---|---|
| **Sub-Strand** | Sub-Strand 1.2: The Atom |
| **Driving Question** | DRIVING QUESTION: Why do different substances produce different colours when burned in a flame, and what does this reveal about how electrons are arranged inside an atom? |

| INSTRUCTIONS |
|---|
| FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson. |

| Lesson # and Title | What did I observe? | What did I learn? | How does this explain the phenomenon? |
|---|---|---|---|
| **Lesson 1**<br>**Anchoring Phenomenon: Flame Colours and the Mystery of the Atom** | Students observed (live demo or video) different metal salts — such as copper(II) chloride, sodium chloride, potassium chloride, and lithium chloride — being held in a Bunsen burner flame. Each substance produced a distinctly different, vivid colour: sodium → bright yellow, copper → blue-green, potassium → lilac, lithium → crimson red. Teacher note: Record all student observations on the class Driving Question Board (DQB); do not correct or validate at this stage — the goal is to surface prior conceptions and generate genuine curiosity. Expect students to note that 'the colour changes depending on what you burn' but to have no explanation for why. | Students learned that the colour produced in a flame is a repeatable, characteristic property of each element — not of the compound or the flame itself. Burning NaCl and NaOH both produce the same yellow colour, pointing to sodium as the source. This is the anchoring phenomenon for the entire unit: something inside each atom must be responsible for the specific colour, but students do not yet have the tools to explain it. Teacher note: Emphasise that this is a genuine scientific puzzle — even trained chemists could not explain flame colours until atomic theory was developed. This builds epistemic motivation for the lessons ahead. | Students cannot yet fully explain the phenomenon — and that is intentional. They can offer initial hypotheses (e.g., 'maybe different substances release different amounts of heat' or 'maybe the colour comes from impurities'). Teacher note: Capture these on the DQB as 'Our Current Best Ideas.' Return to and revise these ideas at the end of every subsequent lesson. The unresolved tension — we can observe a pattern but cannot yet explain it — is the engine that drives inquiry throughout the unit. |
| **Lesson 2**<br>**What Is an Atom? Reviewing Atomic Structure — Dalton & Rutherford Models** | Students examined diagrams and descriptions of Dalton's solid-sphere model and Rutherford's nuclear model, and reviewed the evidence from Rutherford's gold-foil experiment (alpha particles fired at a thin gold sheet — most passed straight through, but a small number were deflected at large angles, and a very few bounced back). Teacher note: Ask students to sketch | Students learned that atoms are not solid, indivisible spheres (Dalton) but instead consist of a tiny, dense, positively charged nucleus — containing protons (positive charge) and neutrons (no charge) — surrounded by electrons (negative charge) moving in a vast region of mostly empty space. Key facts: the nucleus holds virtually all the atom's mass; electrons are negligibly | Students can now partially connect to the driving question: if atoms are mostly empty space with electrons on the outside, then the electrons — being the outermost part of the atom — are most likely what interact with the flame's energy and produce colour. Teacher note: Guide students to update the DQB: 'We now know electrons surround the nucleus — could electrons be responsible |

| | | | |
|---|---|---|---|
| | both models side by side and annotate what each model can and cannot explain. Cold-call at least three students: 'What did Rutherford's result tell us about where the mass of the atom is concentrated?' Wait 10 seconds before accepting answers. | light; the atom is predominantly empty space. Teacher note: Reinforce scale — if the nucleus were the size of a mandazi at the centre of Nairobi's Uhuru Park, the electrons would be orbiting at the park's boundary. This analogy makes the emptiness of atoms viscerally real for students. | for the colours?' This keeps the phenomenon anchored throughout the lesson. Students cannot yet explain why different elements produce different colours, but the focus has shifted meaningfully onto electrons. |
| **Lesson 3**<br>**Bohr's Model and Energy Levels** | Students examined the line emission spectra of hydrogen (and optionally sodium) — either through a spectroscope, a printed spectrum card, or a simulation. They observed that hydrogen does not produce a continuous rainbow of colour but instead emits only specific, discrete lines of colour (e.g., red at 656 nm, blue-green at 486 nm, violet at 434 nm). Teacher note: If physical spectroscopes are unavailable, a printed hydrogen spectrum alongside a continuous white-light spectrum makes an effective comparison. Ask: 'Why does hydrogen only produce four lines of colour rather than every colour?' Allow 15 seconds of think time before cold-calling two students. | Students learned Bohr's planetary model: electrons can only occupy specific, fixed energy levels (shells) numbered n = 1, 2, 3… outward from the nucleus. Shell capacities follow the $2n^2$ rule (n=1: max 2 electrons; n=2: max 8; n=3: max 8 for elements 1–20). Electrons cannot exist between shells. When an electron absorbs energy (e.g., from a flame), it jumps to a higher shell (excited state). When it falls back to a lower shell, it releases a precise packet of energy as light — the colour of that light depends on the size of the energy gap. Because each element has a unique arrangement of shells and electrons, each element's transitions produce a unique set of colours. Teacher note: This is the central explanatory lesson for the driving question. Spend extra time here. Use the analogy of a student climbing stairs in Nairobi's KICC building — they can only stand on steps, not between them. | Students can now construct a foundational explanation for the driving question: sodium produces yellow light because its outermost electron, when excited by the flame, makes a specific energy-level transition that corresponds exactly to yellow light (~589 nm). Potassium's electron makes a different transition → lilac; copper's electrons make different transitions → blue-green. Different elements = different electron arrangements = different energy gaps = different colours of light emitted. Teacher note: Have students write this explanation in their own words and share with a partner before updating the DQB. Check that students distinguish between absorbing energy (jumping up) and emitting light (falling down). |
| **Lesson 4**<br>**Isotopes and Relative Atomic Mass — Part 1: Meaning and Concept** | Students examined data tables showing the naturally occurring isotopes of chlorine (Cl-35 and Cl-37) and carbon (C-12, C-13, C-14), noting that isotopes share the same atomic number (same number of protons and electrons) but differ in mass number (different numbers of neutrons). Teacher note: Use the periodic table to show that chlorine's listed atomic mass (~35.5) is not a whole number — ask students why, and give 20 seconds of wait time. This creates productive confusion that the lesson then resolves. Reference carbon-14 dating, which has been used in East African archaeological sites such as those in the Rift Valley, to make isotopes feel relevant. | Students learned that isotopes are atoms of the same element with the same atomic number but different mass numbers. Because atomic number = number of protons = identity of the element, isotopes are chemically identical (same electron arrangement, same chemical behaviour) but differ in mass. The mass number equals protons + neutrons; so isotopes differ only in neutron count. Teacher note: Stress that isotopes have identical electron configurations — this is critical because electron configuration determines chemical behaviour and, as students will connect to the driving question, also determines the colours produced in a flame. Isotopes of the | Students can now explain why chlorine's atomic mass on the periodic table is 35.5 (not a whole number): it is because naturally occurring chlorine is a mixture of Cl-35 (~75%) and Cl-37 (~25%), and the listed mass is an average of both. Students should also be able to explain why isotopes of the same element produce identical flame colours: their electron arrangements are the same, so the energy-level transitions — and therefore the light emitted — are identical. Teacher note: Return briefly to the driving question: 'Would Cl-35 and Cl-37 produce the same or different flame colours? Why?' This reinforces that electron arrangement, not mass, governs |

| | | | |
|---|---|---|---|
| | | same element produce the same flame colour. | flame colour. |
| **Lesson 5** **Isotopes and Relative Atomic Mass — Part 2: Calculation and Weighted Averages** | Students worked through worked examples and practice problems calculating the relative atomic mass (RAM) of elements from isotope abundance data. For example: chlorine has two isotopes — Cl-35 (abundance 75.77%) and Cl-37 (abundance 24.23%). Students applied the formula: RAM = Σ(isotope mass × fractional abundance). Teacher note: Before revealing the formula, ask students to suggest how they might calculate an 'average' that accounts for the fact that Cl-35 is three times more common than Cl-37. This elicits the weighted-average concept intuitively. Use a relatable analogy: if 3 out of 4 mandazi at a market cost KSh 10 and 1 out of 4 costs KSh 12, what is the average cost? | Students learned that relative atomic mass (RAM) is a weighted average of all naturally occurring isotope masses, calculated using RAM = Σ(isotope mass × fractional abundance). For chlorine: RAM = (35 × 0.7577) + (37 × 0.2423) = 26.52 + 8.97 = 35.49 ≈ 35.5. Students also learned that RAM values on the periodic table are never exact whole numbers for elements with multiple naturally occurring isotopes. Teacher note: Ensure students can distinguish between mass number (whole number, specific isotope) and relative atomic mass (decimal, averaged over all isotopes). A common misconception is that the periodic table shows the mass of a single atom — clarify this explicitly. | Students can now read and interpret the periodic table entry for any element, understanding that the decimal atomic mass reflects the natural isotopic mixture found on Earth. They can calculate RAM from isotope data and explain why two elements with similar mass numbers can have very different chemical properties if their atomic numbers differ. Teacher note: Connect back to the driving question as a brief consolidation: 'Does the RAM of an element affect its flame colour?' (Answer: No — flame colour depends on electron arrangement, which is determined by atomic number, not mass. This is a useful discriminating question for assessment.) |
| **Lesson 6** **From Energy Levels to Orbitals: Introducing s and p Orbitals** | Students examined diagrams of s and p orbitals — the spherical 1s, 2s, 3s shapes and the dumbbell-shaped 2p, 3p orbitals (px, py, pz). They compared the Bohr shell model (simple circular orbits) with the modern quantum mechanical model (probability regions / orbitals). Teacher note: Use a concrete analogy — a Bohr shell is like knowing a matatu operates on a specific route; an orbital is like knowing the probability of where that matatu is at any given moment along the route. Emphasise that orbitals are regions of high electron probability, not defined circular paths. Ask students: 'How is this model an improvement on Bohr's model?' Wait 15 seconds. | Students learned that within each principal energy level (shell), electrons occupy sub-regions called orbitals. The s sub-level contains one spherical orbital (holds max 2 electrons). The p sub-level contains three dumbbell-shaped orbitals (px, py, pz), each holding max 2 electrons (max 6 electrons total in a p sub-level). Principal energy level 1 has only an s sub-level (max 2e); level 2 has s and p sub-levels (max 8e); level 3 has s and p sub-levels for elements 1–20 (max 8e). Teacher note: This lesson upgrades the Bohr model but does not invalidate it — stress that Bohr's core insight (discrete energy levels, electron transitions = light emission) remains correct. The orbital model adds finer resolution. | Students can now explain that when scientists say an electron 'jumps to a higher energy level' in a flame, they mean it moves from one orbital to another of higher energy. The specific orbital-to-orbital transition determines the precise wavelength (colour) of light emitted. Different elements have electrons in different orbitals with different energy gaps — hence different colours. Teacher note: Ask students to update their DQB model sketches to replace simple circular Bohr shells with labelled s and p orbital sub-levels. This is a significant model revision that brings students closer to the full explanation of the driving question. |
| **Lesson 7** **Filling the Orbitals: Aufbau, Pauli, and Hund's Rules** | Students worked through guided examples of filling orbital diagrams (using boxes and arrows for electrons) for elements H through Ne, applying three rules in sequence. They observed what happens when rules are violated — e.g., two electrons with the same spin in one box (Pauli violation) or skipping half-filled | Students learned three rules governing electron configuration: (1) Aufbau Principle — electrons fill the lowest available energy orbital first (1s before 2s before 2p); (2) Pauli Exclusion Principle — each orbital holds a maximum of 2 electrons, and they must have opposite spins (↑↓); (3) Hund's Rule — when filling orbitals of equal energy | Students can now explain why the electron configurations of different elements are uniquely determined and not arbitrary. The specific filling order produces the unique electron arrangement of each element, which in turn produces the unique set of orbital-to-orbital transitions when electrons are excited by a flame. This is why sodium |

| | | | |
|---|---|---|---|
| | orbitals (Hund's violation) — and discussed why these arrangements are less stable. Teacher note: Have students use mini-whiteboards or paper to draw orbital box diagrams for carbon (Z=6) and nitrogen (Z=7) before instruction, then compare with the correct Aufbau/Hund's arrangement. This predict-then-correct structure surfaces and fixes misconceptions efficiently. | (e.g., the three 2p orbitals), electrons occupy each orbital singly with parallel spins before any pairing occurs. Teacher note: Hund's Rule is the most commonly misapplied. Use the analogy of students choosing seats on a matatu — each person takes their own seat before anyone shares. Require students to show spin arrows in orbital diagrams, not just electron counts. | (with its specific filling up to $3s^1$) always produces yellow — the $3s \rightarrow 2p$ transition gap always corresponds to ~589 nm. Teacher note: Connect explicitly to the driving question: 'The rules we learned today are why every sodium atom in the universe produces the same yellow colour — the rules are universal, so the transitions are universal, so the colour is universal.' This is a powerful consolidating insight. |
| **Lesson 8**<br>**Electron Configuration for Elements 1–20 (s and p Notation)** | Students practised writing full electron configurations in s and p notation for all elements from hydrogen (H, Z=1) to calcium (Ca, Z=20), using the Aufbau filling order. They also practised converting between shell notation (e.g., 2,8,1 for sodium) and s/p notation ($1s^2\ 2s^2\ 2p^6\ 3s^1$) and interpreting configurations to identify the element. Teacher note: Include a Kenya-relevant context — for example, note that calcium (found in milk, bones, and Kenyan dairy products from regions like Eldoret and Nyandarua) has the configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$, and iron (found in sukuma wiki and beans, important for nutrition) has a configuration that extends beyond this course's scope but follows the same rules. | Students learned to write and interpret full s/p electron configurations for elements 1–20. Key configurations to master: H: $1s^1$; He: $1s^2$; Li: $1s^2\ 2s^1$; C: $1s^2\ 2s^2\ 2p^2$; N: $1s^2\ 2s^2\ 2p^3$; O: $1s^2\ 2s^2\ 2p^4$; Ne: $1s^2\ 2s^2\ 2p^6$; Na: $1s^2\ 2s^2\ 2p^6\ 3s^1$; Mg: $1s^2\ 2s^2\ 2p^6\ 3s^2$; Al: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$; Si: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$; Cl: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$; Ar: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$; K: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$; Ca: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$. Teacher note: Stress that the outermost (valence) electrons are the ones that participate in flame excitation. Group 1 elements (Li, Na, K) each have a single s electron in their outermost shell — this is why they are particularly easy to excite and produce vivid, clean flame colours. | Students can now give a complete, technically precise explanation of why different elements produce different flame colours: each element has a unique electron configuration (determined by atomic number and the Aufbau/Pauli/Hund's rules); when heated in a flame, outer electrons absorb energy and jump to higher orbitals; when they return to the ground state, they emit photons whose energy — and therefore colour — corresponds exactly to the energy gap of that specific transition; because each element's orbital energy gaps are unique, each element emits a characteristic colour or set of colours. Teacher note: This is the penultimate lesson before the full DQB resolution. Ensure all students can write the configuration for Na and K and connect it to the yellow and lilac flame colours observed in Lesson 1. |
| **Lesson 9**<br>**Model Building, DQB Resolution, and Unit Consolidation: Answering the Driving Question** | Students returned to the original flame-colour observations from Lesson 1 and to every entry on the Driving Question Board accumulated over the unit. They reviewed the progression of atomic models — Dalton (solid sphere) → Rutherford (nuclear model) → Bohr (energy levels/shells) → Modern orbital model (s/p orbitals, Aufbau/Pauli/Hund's) — and identified the specific experimental evidence that forced each revision. Teacher note: Display all four model diagrams side by side. Ask students: 'What evidence made scientists reject each earlier model?' Cold-call four students (one per model transition). Give 20 seconds of | Students learned that the history of atomic theory is a story of models being revised — not discarded as failures, but refined as new evidence emerged. Each model was the best explanation available at the time, and each revision made atomic theory more powerful and more precise. The final s/p orbital model now allows chemists to predict electron configurations, chemical bonding, reactivity, and — returning to the start — the specific colours elements emit in a flame. Students also consolidated understanding that science progresses through cycles of observation, modelling, testing, and revision. Teacher note: This | Students can now construct a complete, multi-part explanation for the driving question: (1) Atoms consist of a nucleus (protons + neutrons) surrounded by electrons in specific orbitals, arranged according to the Aufbau, Pauli, and Hund's rules. (2) Each element has a unique atomic number → unique electron configuration → unique set of orbital energy levels. (3) When an element is placed in a flame, outer electrons absorb thermal energy and are promoted to higher-energy orbitals (excited state). (4) As electrons fall back to the ground state, they emit photons of light; the colour of the photon |

| | | | |
|---|---|---|---|
| | think time. Expect students to cite: Rutherford's gold-foil experiment; hydrogen line spectra (for Bohr); and the limitations of Bohr's model for multi-electron atoms (for the orbital model). | metacognitive lesson on the nature of science is as important as the content. Ask students: 'Is the orbital model the final, perfect model of the atom?' Guide them to appreciate that further refinements (d and f orbitals, quantum numbers) exist beyond Grade 10, and that science is always open to revision. | corresponds to the energy of the specific orbital transition. (5) Because each element's orbital energy gaps are unique, each element emits a characteristic colour — sodium always emits yellow (~589 nm) because its 3s→ground-state transition always releases the same energy. This is why flame tests can identify elements. Teacher note: Have every student write this explanation independently and in full sentences before the end of the lesson. This is the culminating assessment of the unit. Compare student explanations to their Lesson 1 initial ideas on the DQB — this growth in understanding is the most powerful feedback tool available. |